

2014 F=ma Exam: Problem 3

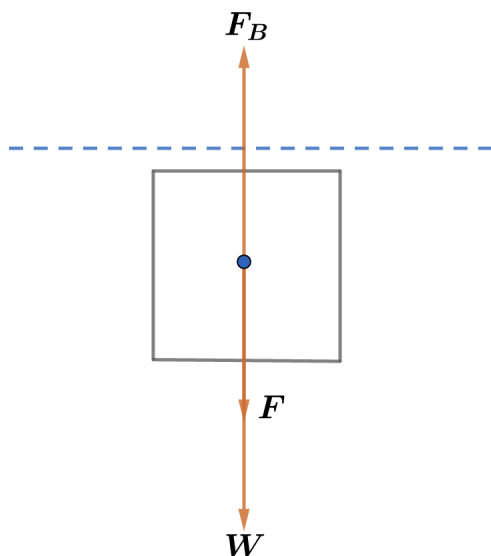
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Recall the fraction of an object submerged in water is given by

$$f_{\text{sub}} = \frac{\rho_{\text{obj}}}{\rho_{\text{water}}}$$

In our case,

$$\begin{aligned} f_{\text{sub}} &= 1 - f_{\text{above}} = 1 - 0.2 = 0.8 \\ \rho_{\text{obj}} &= 0.8\rho_{\text{water}} \end{aligned}$$



When the external force $F = 3 \text{ N}$ acts on the object, it becomes fully submerged. Balancing forces on the object,

$$\begin{aligned} F_B &= F + W \\ \rho_{\text{water}} V g &= F + \rho_{\text{obj}} V g \end{aligned}$$

Solving for the volume V ,

$$(\rho_{\text{water}} - \rho_{\text{obj}}) V g = F$$

Since $\rho_{\text{obj}} = 0.8\rho_{\text{water}}$,

$$V = \frac{F}{(\rho_{\text{water}} - \rho_{\text{obj}})g} = \frac{F}{0.2\rho_{\text{water}}g} = \frac{3 \text{ N}}{0.2(1000 \text{ kg/m}^3)(10 \text{ m/s}^2)} = 0.0015 \text{ m}^3 = 1.5 \text{ L}$$

so the answer is D.